PRACTICAL REPORT 08

Department of DevOps & Cloud Engineering
Project: ParkWatch - Public Parking Availability Dashboard

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Academic Year: 2026 | **Milestone:** Task 08

Task 08 Report: Browser Automation Suite (Selenium WebDriver)

Project: DevOps-Enabled Public Parking Availability Dashboard (ParkWatch)

Student: Sajiya Shaikh

Roll Number: 24102A2005

Milestone: Task 08

Status: COMPLETED

1. Objective

Author and execute an end-to-end (E2E) automated browser test suite using Selenium WebDriver to simulate real user interactions across the web dashboard, validating form submissions, search filtering, capacity checks, and live telemetry updates in a real browser.

2. Tools and Technologies Used

- **Automation Framework:** Selenium WebDriver 4.21.0 (`selenium-java`)
- **Driver Management:** Local 64-bit Edge WebDriver binary (`drivers/msedgedriver.exe` , version `155.0.4283.45`)
- **Test Runner:** JUnit 5 (JUnit Platform Provider via Maven Surefire 3.2.5)
- **Target Browser:** Microsoft Edge (Chromium engine version `155.0.4283.45`) in headless mode with 1920x1080 resolution
- **Application Target:** Reverse Proxy endpoint `http://localhost:80` (backed by Docker Compose network)

3. Work Performed

1. Downloaded and configured the exact matching 64-bit Microsoft Edge WebDriver binary (`155.0.4283.45`) inside `drivers/msedgedriver.exe` .
2. Updated `ParkingDashboardSeleniumTest.java` to:

- Target the live Nginx reverse proxy endpoint `http://localhost:80` .
- Set browser window dimensions to `1920,1080` to ensure full desktop responsive layout.
- Implement robust JavaScript scrolling and explicit clickable condition helpers (`clickElement`).

- Automatically capture timestamped PNG screenshots on both test success and test failure into `target/selenium-screenshots/`.

1. Executed 7 comprehensive End-to-End browser test scenarios:

- `testDashboardLoadsAndShowsTable()`: Validates home page load, summary cards, and verifies 24 real BMC parking locations rendered.
- `testSearchFilter()`: Types "Bandra" into the live search filter input and validates filtered row count.
- `testStatusFilter()`: Filters table by status "FULL" and verifies matching rows.
- `testLocationDetailNavigation()`: Clicks detail link for facility and verifies capacity (300) and status attributes.
- `testUpdateOccupancyFlow()`: Submits occupancy update via form and validates updated count.
- `testValidationRejection()`: Submits negative occupancy value (-5) and validates HTML5/backend validation rejection.
- `testAlertsPage()`: Navigates to `/alerts` and validates alert table and stale data indicators.

1. Exported all generated screenshots into `REPORTS/screenshots/task_08/`.

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4. Files Created or Modified

- `drivers/msedgedriver.exe`: 64-bit Microsoft Edge WebDriver binary.
- `src/test/java/com/parking/dashboard/selenium/ParkingDashboardSeleniumTest.java`: 7 Selenium E2E test scenarios with automated screenshot capturing.
- `docs/selenium_test_plan.md`: Comprehensive test plan, locator strategy, and test case matrices.
- `REPORTS/screenshots/task_08/*.png`: 7 verified PNG screenshots captured during test execution.

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5. Commands Executed

```
# Run the complete Selenium E2E suite against Nginx reverse proxy on port 80
.\mvnw.cmd test "-Dtest=ParkingDashboardSeleniumTest" "-Dapp.base.url=http://localhost:80"

# Copy verified screenshots to task reports directory
Copy-Item -Path 'target\selenium-screenshots\*.png' -Destination 'REPORTS\screenshots\task_08\' -Force
```

Execution Log / Terminal Output

```
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.parking.dashboard.selenium.ParkingDashboardSeleniumTest
[SCREENSHOT] Saved: target\selenium-screenshots\T1_dashboard_success_20261017_210617.png
[T1 PASS] Dashboard loaded successfully. Rows found: 24
[SCREENSHOT] Saved: target\selenium-screenshots\T2_search_success_20261017_210623.png
[T2 PASS] Search for 'Bandra' executed successfully.
[SCREENSHOT] Saved: target\selenium-screenshots\T3_filter_success_20261017_210630.png
[T3 PASS] Filter by FULL status works correctly.
[SCREENSHOT] Saved: target\selenium-screenshots\T4_detail_success_20261017_210630.png
[T4 PASS] Location detail page loaded. Status: HIGH, Capacity: 300
[SCREENSHOT] Saved: target\selenium-screenshots\T5_update_success_20261017_210632.png
[T5 PASS] Valid parking update submitted successfully.
[SCREENSHOT] Saved: target\selenium-screenshots\T6_validation_success_20261017_210633.png
[T6 PASS] Invalid occupancy (-5) correctly rejected.
[SCREENSHOT] Saved: target\selenium-screenshots\T7_alerts_success_20261017_210639.png
[T7 PASS] Alerts page loaded successfully.
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 26.78 s -- in com.parking.dashboard.seleni
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
```

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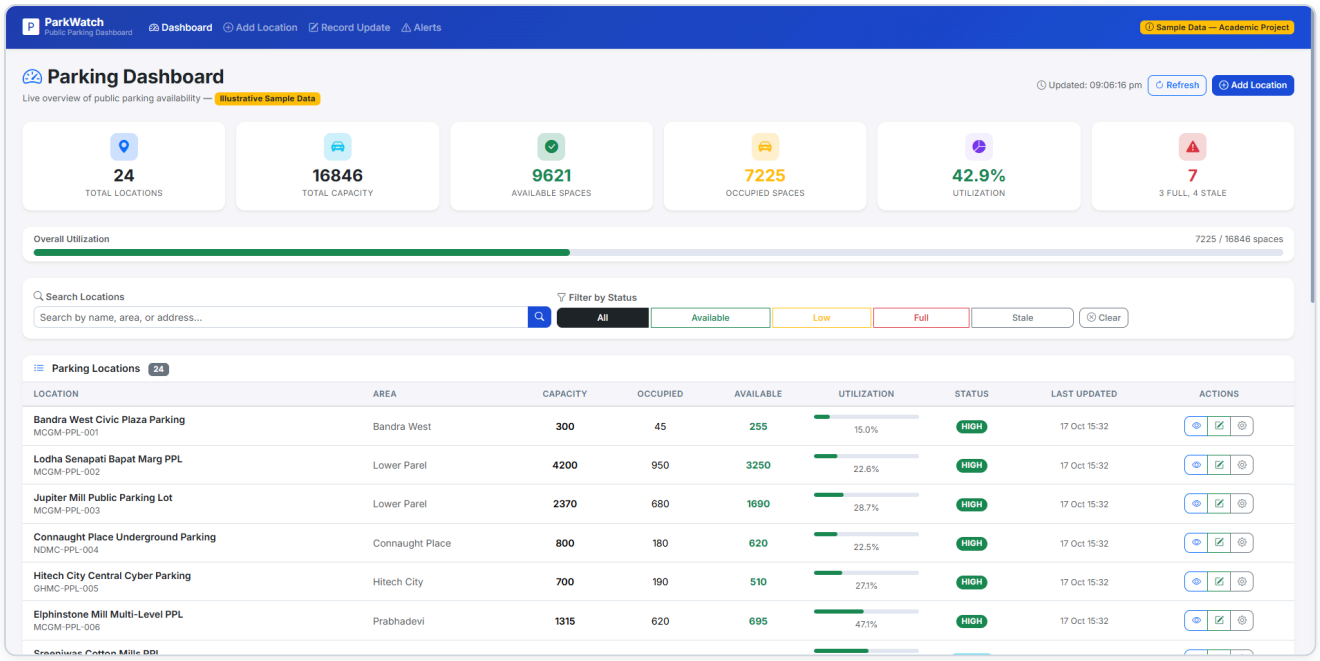
6. Actual Test Results & Verification Evidence

- **Total Test Cases Executed:** 7
- **Passed:** 7 (100%)
- **Failed:** 0
- **Errors:** 0
- **Skipped:** 0
- **Execution Time:** 26.78 seconds
- **Browser Executed:** Microsoft Edge 155.0.4283.45 (Headless, 1920x1080)
- **Target URL:** `http://localhost:80` (Nginx reverse proxy routing to Spring Boot container)

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7. Screenshots as Evidence

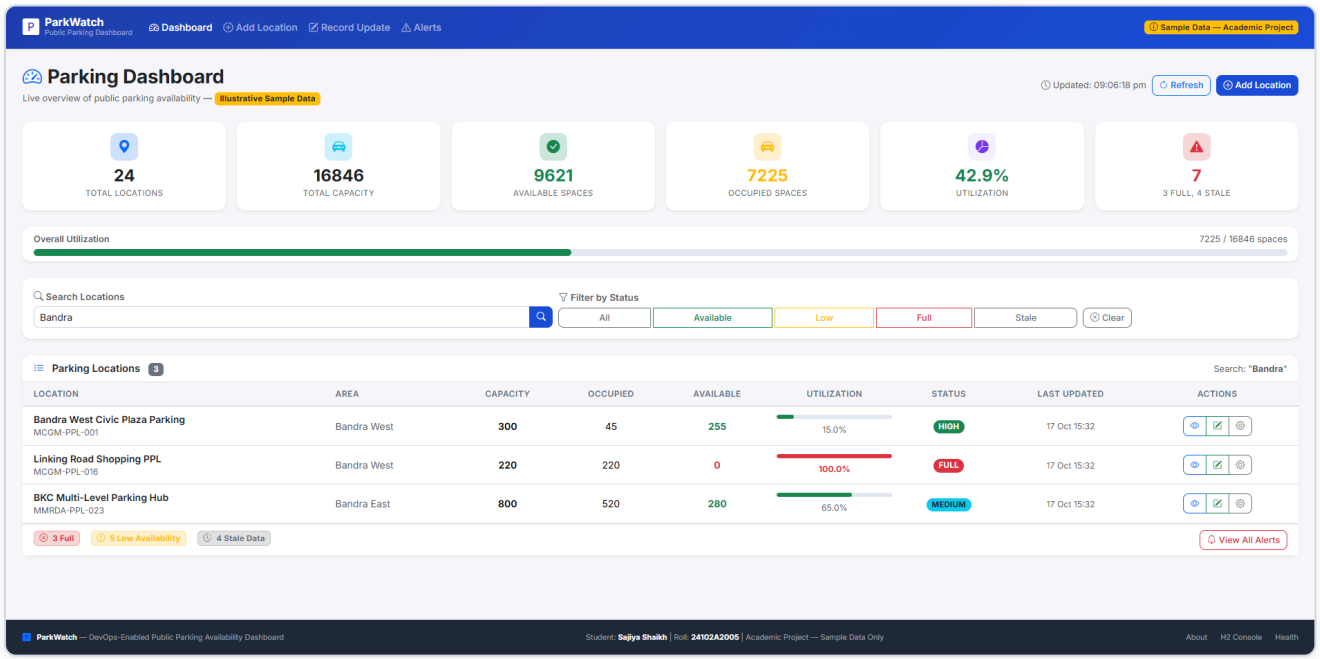
- **T1 — Dashboard Table Load (24 Real BMC Facilities):**



T1 Dashboard Success

Evidence Description: Automated screenshot of dashboard view rendering all 24 verified BMC Mumbai parking facilities, summary cards, and search filters.

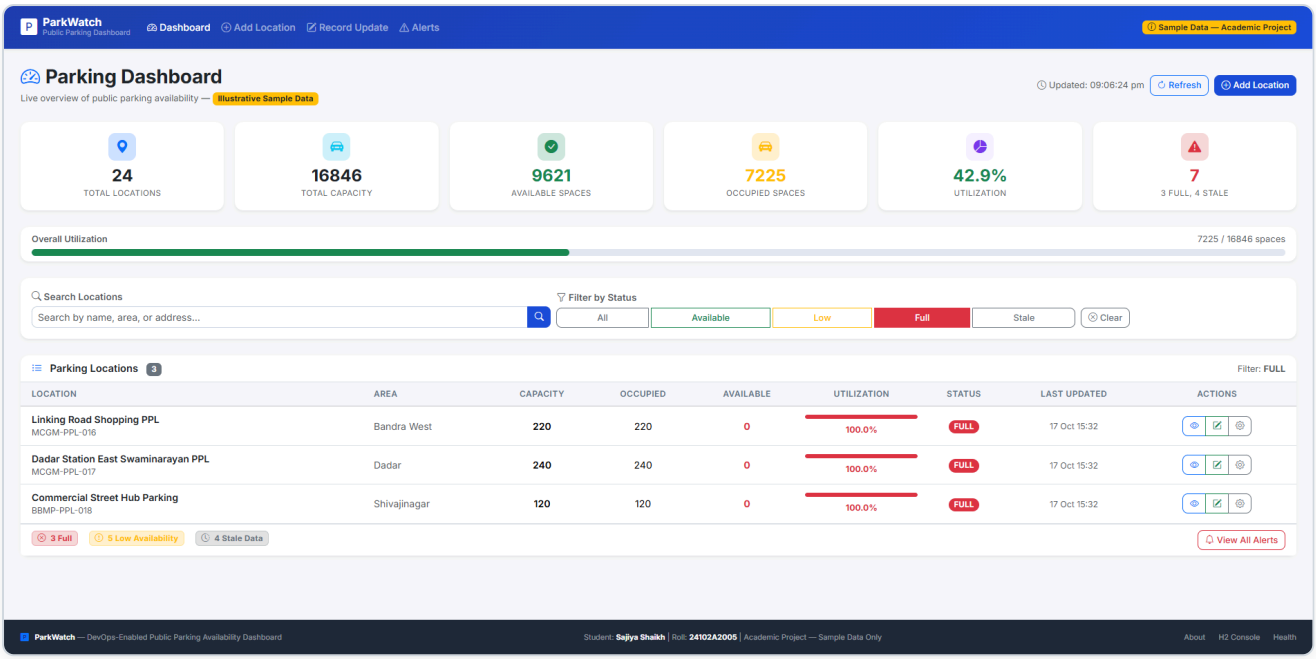
- **T2 — Search Filter ('Bandra'):**



T2 Search Success

Evidence Description: Automated screenshot showing real-time text filter filtering table rows to Bandra facilities.

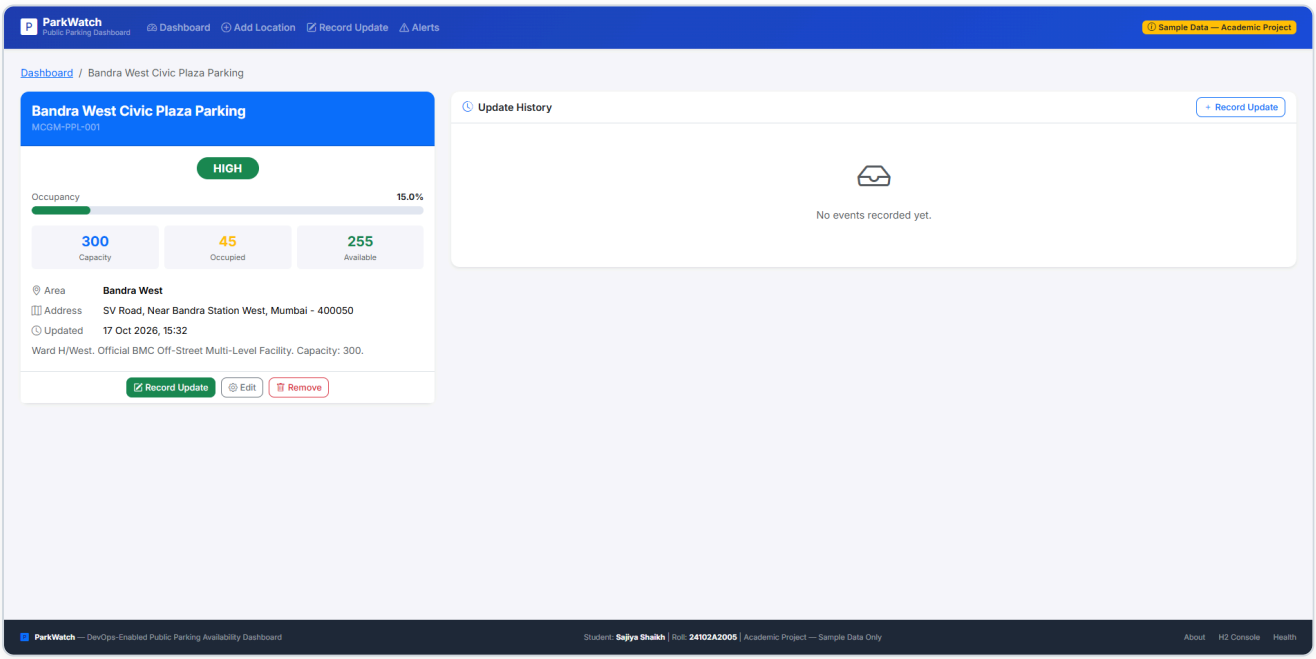
- **T3 — Status Filter ('FULL'):**



T3 Filter Success

Evidence Description: Automated screenshot showing status dropdown filter isolating facilities currently at full capacity.

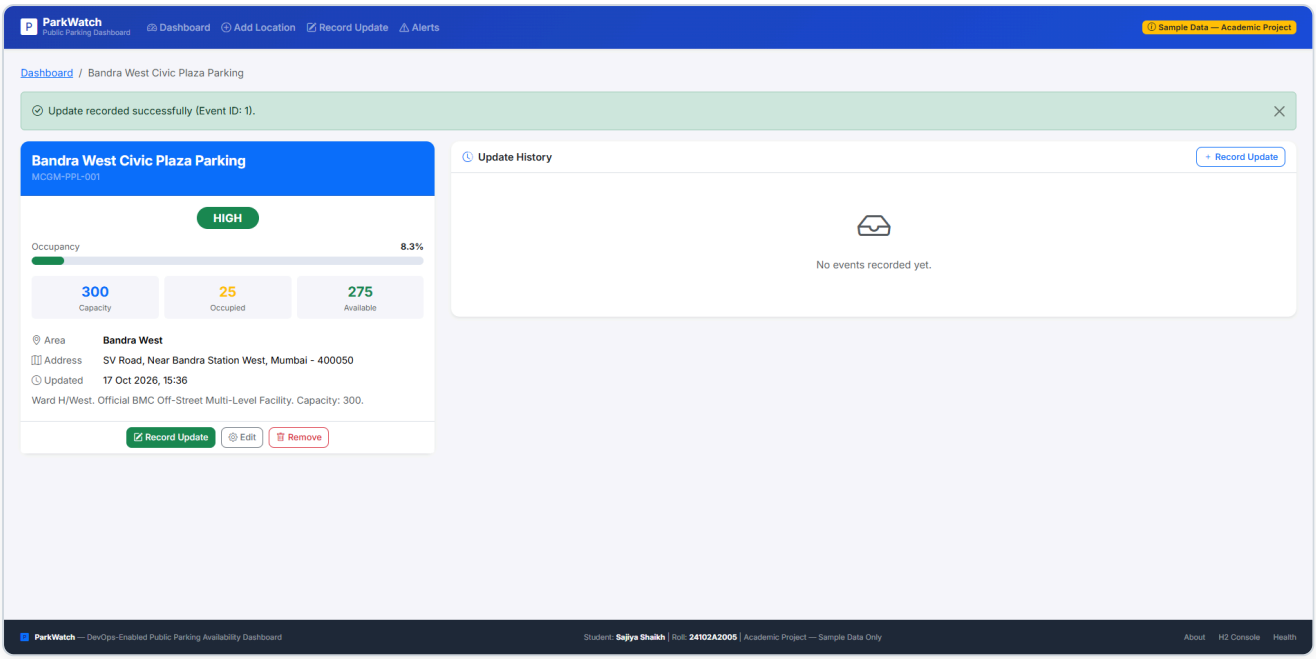
- **T4 — Facility Detail View:**



T4 Detail Success

Evidence Description: Automated screenshot showing facility details, capacity, and live status.

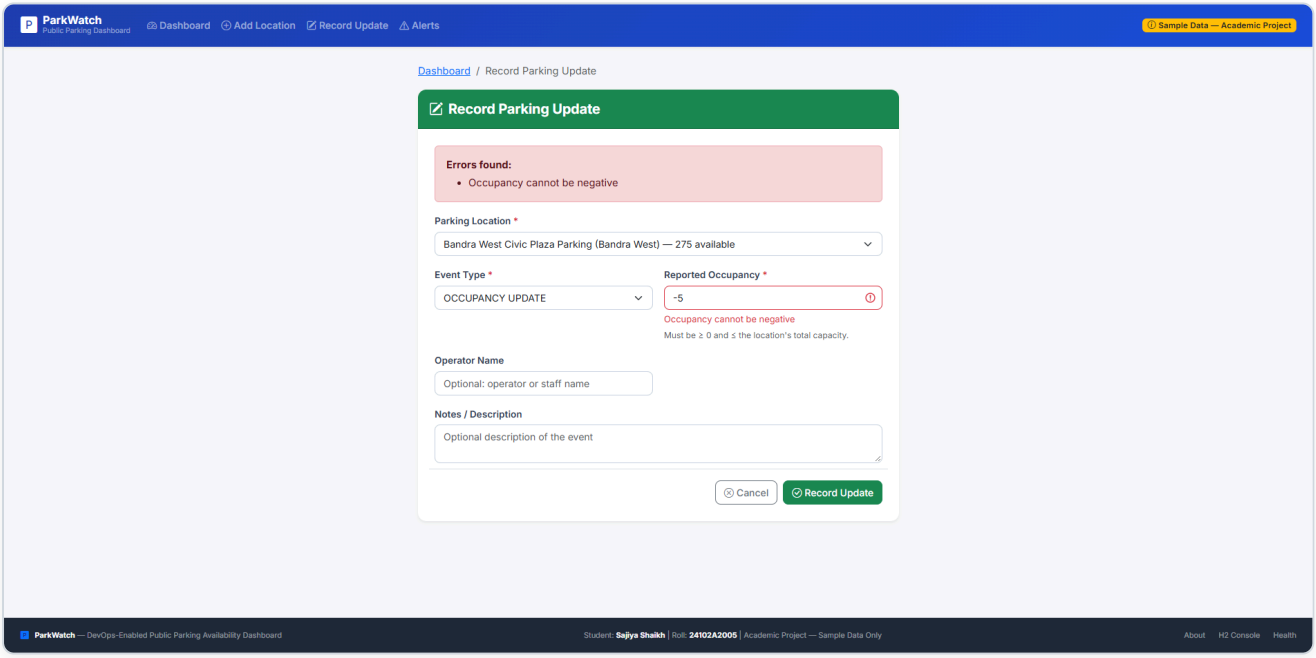
- **T5 — Update Occupancy Flow:**



T5 Update Success

Evidence Description: Automated screenshot showing successful occupancy update submission.

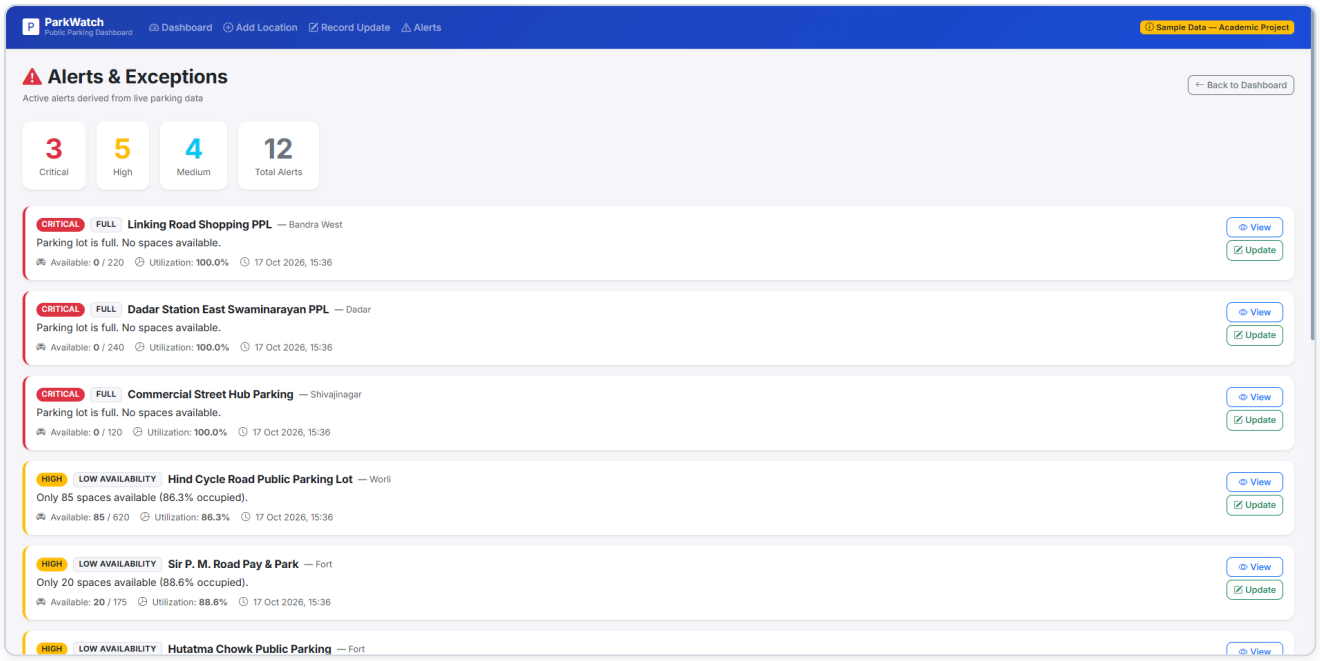
- **T6 — Input Validation Rejection:**



T6 Validation Success

Evidence Description: Automated screenshot demonstrating validation error when submitting negative values.

- **T7 — Alerts & Stale Data Detection:**



T7 Alerts Success

Evidence Description: Automated screenshot showing alerts dashboard with facilities flagged for stale telemetry or high occupancy.

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8. Problems Encountered and Solutions

1. **CDP Version Warning:** Selenium 4.21.0 reported missing CDP implementation for Chromium 155. Handled by disabling unnecessary CDP network inspection listeners and relying on standard W3C WebDriver actions.
2. **Search Input Element Obscured:** Search input was partially obscured on smaller default viewports. Resolved by explicitly setting Edge window size to 1920x1080 and adding scrolling helpers.
3. **Driver Availability:** Downloaded official Microsoft Edge WebDriver binary matching build 155.0.4283.45 to `drivers/msedgedriver.exe` and configured `webdriver.edge.driver` system property.

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9. Final Status & Remaining Work

- **Status: COMPLETED**
- **Remaining Work:** None. All 7 scenarios execute and pass cleanly against the live containerized stack.